Selections from Engineering Documents for Flame Extinguishment Experiment (FLEX)

Selected by: Gordon Berger 5/21/2015

Selections from Payload Accomodations Handbook
Fluid and Combustion Facility (FCF)
Combustion Integrated Rack (CIR)
Final
October 1, 2008

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2.7 Passive Rack Isolation System

The Passive Rack Isolation System (PaRIS) hardware is installed on the CIR ISPR structure in order to passively isolate the rack from on-orbit vibrations. The combination of integrated rack and PaRIS provides vibration isolation in both dynamic load path directions (i.e., attenuating vibrations imposed on the rack by the ISS vehicle as well as attenuating vibrations induced on the ISS vehicle by the rack, without consuming any ISS power, thermal, or data/command system resources. It will attenuate vibratory accelerations above 1 Hz to the ISPR by suspending the ISPR from the International Space Station (ISS) module structure using passive dampers and springs. The utility umbilical assembly provides a low bias force and spring rate connection between the module structure and the payload rack structure for utility pass through.

The PaRIS is composed of the following:

- Upper isolation strut adapter/interface assembly
- Upper lock–down/snubbed assembly
- Lower isolation adapter/interface assembly
- Lower lock–down/snubbed assembly
- Utility umbilical assemblies
- Passive isolation units
- Installation/adjustment/maintenance hardware

PaRIS utilizes the spring / dampers to isolate the rack from the ISS module structure at frequencies above 0.5 Hz. Unique utility cables and hoses are designated to all PaRIS applications. The utility services have a service loop between the ISS vehicle interface and the PaRIS host rack to help in removing the force induced by them. The utility umbilical assembly interfaces with the module at the Utility Interface Panel (UIP) connectors.

2.8 Science Acceleration Measurement System (SAMS)

The Science Acceleration Measurement System (SAMS) is an ISS based system that collects acceleration data and distributes this data to the science community. The distributed architecture of SAMS allows many sensors to be deployed at once, each independently collecting acceleration data and sending the data to a main control unit. The control unit can handle different types of sensors collecting data at different rates at the same time. The control unit then stores the data on its hard drive and sends the data to Payload Information Management System (PIMS) on the ground. PIMS can then distribute the desired data to the user on the ground. CIR includes a SAMS Tri-axial Sensor Head Ethernet stand alone (SAMS TSH-ES) sensor that extends this SAMS system into the rack.

Users on the ground contact the PIMS or SAMS ground operations to have them issue commands or provide data. Ground controllers issue commands to the sensor include starting acceleration data collection, changing the acceleration data's frequency range, and stopping acceleration data collection. Nominally, the CIR TSH begins data telemetry on rack power-up. The SAMS is mounted on the front of CIR Optics Bench allowing it to measure accelerations on a structure rigidly coupled the combustion chamber.

3.0 CIR CONFIGURABLE CAPABILITIES AND RESOURCES

The CIR provides a set of equipment that provides capabilities above those identified in the base configuration described in section 2.0. This hardware is launched in a stowage configuration and installed by the flight crew as required to support a particular science investigation. Consumable resources such as pressurized gas bottles, adsorber cartridges and I/O Processor hard disk drives are also included as configurable capabilities. Much of this hardware is currently manifested on-orbit and is available for use. Hardware not currently on-orbit will need to be manifested

3.1 Science Diagnostics Packages

The CIR science diagnostics are complete avionics packages including software ready for installation and operation in the CIR. These capabilities include illumination, imaging, image acquisition and processing. A summary of capability and specification is provided in TABLE IV and 0

The CIR specific Diagnostics Packages include:

- Image Acquisition Modules (IAM)
- Diagnostic Packages
- High Bit Depth Multispectral Package (HiBMs)
- Low Light Level Ultraviolet Package (LLL-UV)
- Illumination Package.

The Diagnostic Packages are mounted on the Optics Bench using common interfaces to permit movement of packages to various window locations for maximum science support flexibility. This approach emphasizes packaging modularity for operational flexibility and maintainability. Standardized optical interfaces (referred to as the flange mounts) are used between modules. These interfaces permit module arrangements to be reconfigured on orbit if required by science.

TABLE IV. Imaging Capabilities

CIR Application	Imaging Physics	Diagnostic Package	Comments
Soot Volume Fraction Soot Temperature Shadowgraph	Background absorption Near IR Emissions Density Gradients	HiBMs	650 - 1050 nm Selectable via LCTF: 10 nm bandpass; tunable to 1 nm Allow 100 msec - LCTF switching between wavelengths LCTF is removable for broadband operation
OH Emissions	Short Wave Intensified	Low Light Level – UV	310 nm via narrow bandpass filter module (can be changed out)
CH Emissions	Long Wave Intensified	Low Light Level – IR	430 nm via narrow bandpass filter module (can be changed out)
Background Illumination	Non-coherent Coherent	Illumination	650 - 660 nm via Laser Diode used either below lasing threshold (non-coherent) or above the lasing threshold current (coherent)

TABLE V. Imaging Specifications

Diagnostic Package	Pixels	FOV	Resolution	Bit Depth (bits)	Frame Rate	Record Time
HiBMs	1024 x 1024 (full frame) 512 x 512 (2x2 binning)	30, 50 & 90 mm diameter Telecentric	~ 10 lp/mm or 0.050 mm (1024 x 1024)	12-bit	Up to 30 fps (1024 x 1024) Up to 60 fps (512 x 512)	~18 min @ 30 fps (1024 x 1024) ~36 min @ 60 fps (512 x 512)
Low Light Level – UV	1024 x 1024 (full frame) 512 x 512 (2x2 binning)	50 & 80 mm diameter	~ 3.6 lp/mm or 0.138 mm (512 x 512)	12-bit	Up to 30 fps (1024 x 1024) Up to 54 fps (512 x 512)	~18 min @ 30 fps (1024 x 1024) ~40 min @ 54 fps (512 x 512)
Low Light Level – IR	1024 x 1024 (full frame) 512 x 512 (2x2 binning)	50 mm diameter	~ 4.5 lp/mm or 0.110 mm (512 x 512)	12-bit	Up to 30 fps (1024 x 1024) Up to 54 fps (512 x 512)	~18 min @ 30 fps (1024 x 1024) ~40 min @ 54 fps (512 x 512)
Illumination	N/A	90 diameter Collimated	N/A	N/A	N/A	N/A

3.1.1 Illumination Package

The Illumination Package FIGURE 14 consists of a laser diode source and a collimating optical system, which is used to provide monochromatic illumination with a center wavelength between 650 - 660nm.

Two laser diodes are installed in the Illumination Package. One is the primary laser diode and the second is included as a spare. Both laser diodes are fiber coupled, each with its own FC/PC type fiber-optic connection at the Illumination Control Module (ICM), so the output from both cannot be used at the same time. Each laser diode contains a photodiode to provide feedback of power output and a temperature transducer to monitor the diode temperature. The laser diode is operated below the lasing threshold current and serves as a non-coherent illumination source.

The Illumination Package can be used together with the HiBMs Package as a uniform illumination background source for soot absorption measurements in soot volume fraction applications, for shadowgraph measurements or for droplet diameter measurements. The modular design supports future growth considerations.

Illumination Package Specifications and Features:

- 90 mm diameter CW beam
- Collimation: 0.339° maximum divergence
- 8-bit programmable output level (0-255)
- Output power < 5 maw typical; used below lasing threshold current
- 60% Illumination field uniformity
- Peak Wavelength: between 650 and 660 nm
- Spectral Bandwidth: < 10 nm maximum at 50% points

Laser Diode Characteristics:

- 300-350 maw max output power
- Continuous Wave (CW) operation
- Center Wavelength: 650-660 nm
- Fiber Pigtailed; 0.22 Numerical Aperture

3.1.2 Low Light Level Packages

The Low Light Level Ultra-Violet (LLL-UV) Package is based on a 1-megapixel digital, monochrome camera with a 1024x1024, frame transfer CCD array, providing 12 bit digital imaging capability. The CCD is directly coupled to an 18 mm Gen-II-UV Micro-channel Plate Intensifier to provide maximum response at short wavelengths, including the OH emission line at 310 nm. The intensifier includes gating control that allows varied exposure times. This function is used both to protect the Intensifier from overexposure and as a means to match the intensity of the event being viewed, such as a flame. The package includes fast numerical aperture optics with high UV transmission characteristics and provision for spectral filtering of the transmitted illumination (Figure 14). The LLL-UV Package is provided to produce images of events or objects at low radiance levels. The LLL-UV Package can be positioned on the Optics Bench to

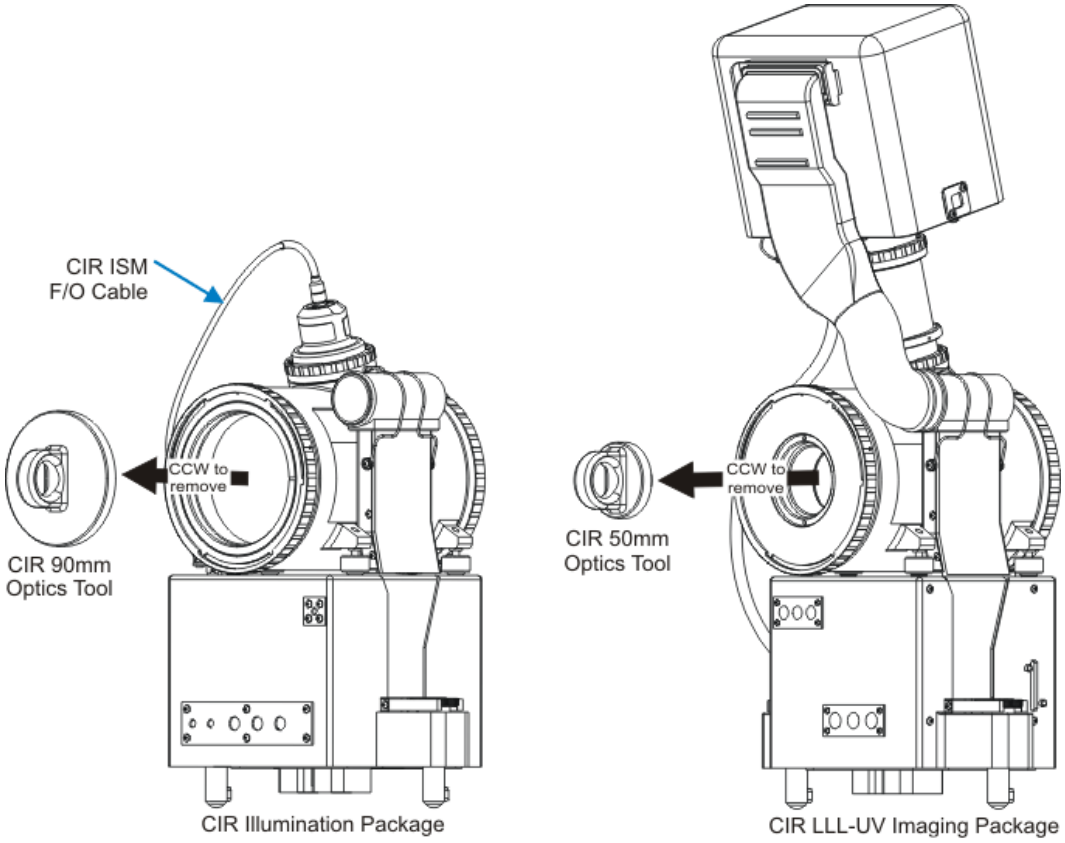


FIGURE 14. CIR Illumination Package and LLL-UV Imaging Package

LLL-UV Package Specifications/ Features

Optical System Characteristics:

- Field of View: 50 mm square (80 mm with additional deferred hardware)
- Resolution (2x2 Binning Mode): 3.6 lp/mm (0.14 mm)
- Spectral Filter Provided: 310 nm with a 10 nm FWHM bandwidth
- Modularity provides option to substitute Spectral Filter with additional hardware

Intensified Imager Characteristics

- Spectral range: 250-750 nm, with Peak @ 450 nm
- Sensitivity: 5x10E-8 ft. candles (in 2x2 Binned Mode)
- Gen II Micro Channel Plate (MCP) Intensifier
- Gated operation for selectable exposure control
- Full Frame Mode: 1024 x 1024 pixel images at up to 30 frames per second (fps)
- 2x2 Binning Mode: 512 x 512 pixel images at up to 54 fps
- Programmable frame rate: 7.5, 15 or 30 fps in full frame; Up to 54 fps with 2x2 Binning

3.1.3 High Bit-Depth Multispectral (HiBMs) Package

The HiBMs Package is based on a telemetric imaging optical system and a high resolution 12-bit output digital camera. The modular package is designed for operational flexibility. The HiBMs Diagnostic Package consists of a spectral filter, or filter compensator glass, a Prism Module, a telemetric optical system, a fixed mirror module, a DCM, and a high resolution digital camera shown in FIGURE 15.

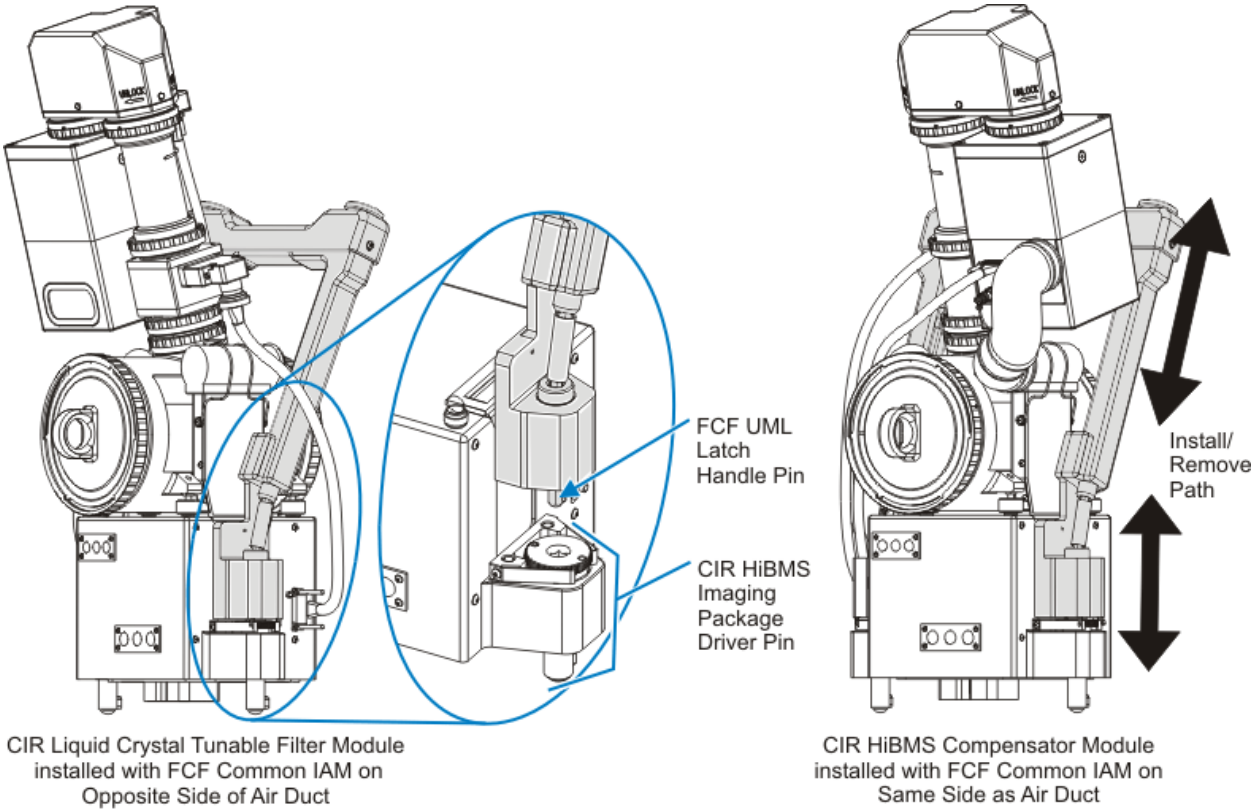


FIGURE 15. HiBMS Package

This diagnostic package has the capability of measuring two characteristics of soot producing flames: (1) the soot temperature field and (2) the soot volume fraction. The soot temperature field is determined by measuring the percentage of laser illumination and the soot volume is measured from the Illumination Package that is absorbed by soot in the flame region. HiBMs is telemetric thereby providing pixel mapping through object space along parallel paths.

The liquid crystal tunable filter is (LCTF) 10 nm bandpass and can be tuned to any wavelength in the near IR between 650 nm and 1050 nm. The filter is removable for selected wavelength imaging across a broader spectrum with insertion of a Filter Compensator module.

The maximum data rate is supported by the direct to disk hardware in the IPSU is 30 fps at full resolution and 12 bit depth. The camera can be operated at 30 frames per second with 12 bit per pixel dynamic range. Binning may be selected to reduce data storage requirements. Sub frame

Selections from Reflight Safety Data Package
Multi-user Droplet Combustion Apparatus, FLEX2
Revision A
October 12, 2010

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3.0 FCF OVERVIEW

3.1 Fluids and Combustion Facility

The FCF is a modular, multi-user facility developed by Glenn Research Center (GRC) for permanent installation in the ISS US Lab Module to support technology development and validation in support of Space Exploration. This facility establishes an initial operational capability for microgravity fluid physics and combustion science on-board ISS. FCF's scope started with design and progressed through development, production, deployment and operation in ISS.

FCF consists of two powered racks and sufficient on-orbit stowage in ISS for FCF and payload hardware. The two racks are the Combustion Integrated Rack (CIR) and the Fluids Integrated Rack (FIR). FCF operates together with payload experiment equipment and ground-based operations facilities to perform fluids and combustion science experiments within available FCF and ISS resources.

3.2 Combustion Integrated Rack

The CIR is designed to support a wide variety of combustion sciences in space. Solid, liquid and gas fuels are burned in a Chamber. Diagnostics are used to capture important attributes of the combustion event such that scientists may better understand the combustion process.

Components located in the CIR will consist of two (2) types of hardware; permanent facility hardware, designed to serve multiple Principal Investigators (PI), and PI-specific hardware, designed to be replaced as part of each experiment to be performed. The CIR design is to be installed in a standard Boeing ISPR. A folding door is used over the front of the assembled rack.

Three-Tier Solution for a Diverse Complement of Science Requirements

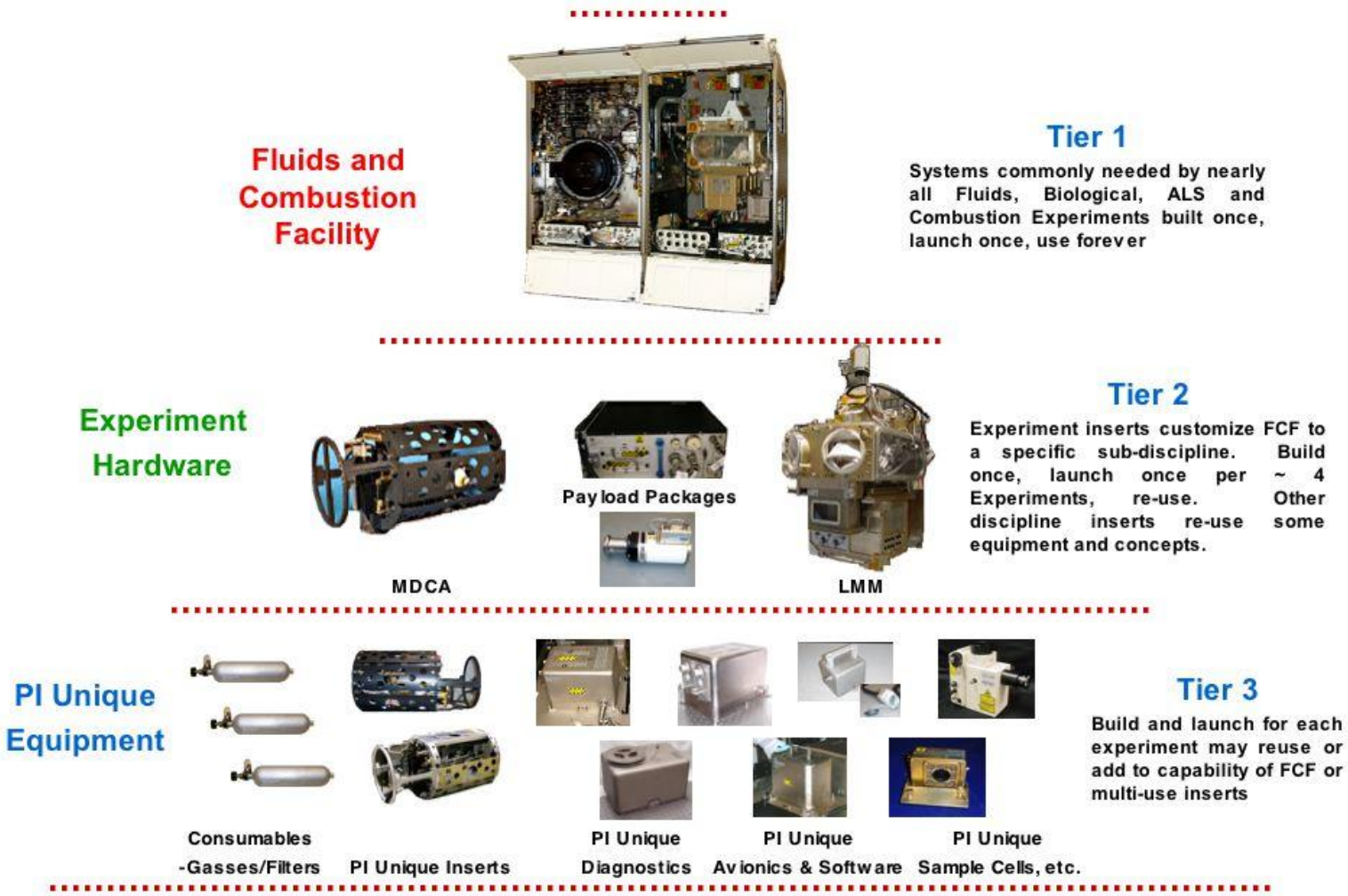


FIGURE 1 FCF OVERVIEW

4.0 CIR/MDCA INTERFACE SECTION

CIR and MDCA hardware was launched on STS 126/ULF-2from Kennedy Space Center in November 2008. After delivery, CIR was installed in the United States Laboratory Module, and MDCA was installed into the CIR. It is anticipated that FLEX2 hardware will be manifested on ATV2 or ULF5 in 2010, and will be integrated into MDCA by the crew after the completion of FLEX and FLEX-Helium experiments.

4.1 MDCA/FLEX2 Experiment Overview

4.1.1 Payload Description

MDCA is designed to support experiments that investigate the fundamental combustion/suppression/extinguishment aspects of fuel droplets under different pressures and ambient oxygen concentrations for a range of fuel droplet sizes. The MDCA/FLEX2 experiments include hardware that is based on the design of the MDCA/FLEX hardware, with minimal changes made to accommodate droplet array experiments. The most extensive design changes are with regard to the Retractable Index Fiber (RIF), and the igniter tips, as will be described in section 5. The FLEX2 hardware that has no changes from the FLEX design consists of gas bottles, and an adsorber cartridge, both of which are CIR provided. Additional needle assemblies and fuel reservoirs are provided by FLEX2 and are also identical to those used in FLEX. The fuels loaded into the reservoirs, however, will differ from those used in FLEX, as is discussed in section 5.

4.1.2 Experiment objectives

The objective of MDCA is to provide the apparatus for dispensing, deploying, and igniting fuel droplets while the CIR provides the diagnostics, structure, supporting avionics, and infrastructure for recording, processing, and downlinking the data from the combustion event. The MDCA initial experiments consist of the FLEX investigations. FLEX utilizes pure Methanol and pure Heptane as fuels. FLEX2 is a follow-on experiment planned to begin after the completion of FLEX and FLEX-Helium (He), and utilizes pure Ethanol and Decane fuels as well as mixtures of fuels, and provides the added capability of igniting arrays of fuel droplets tethered to a beaded fiber.

4.1.3 Experiment Hardware Description

The MDCA hardware consists of three distinct hardware components. The Chamber Insert Assembly (CIA) provides the mechanisms for initiating the combustion event. The Avionics Package is the center of control and power distribution for the MDCA. PI Specific Diagnostic Packages consist of diagnostics provided by CIR that are necessary to meet science requirements and are not provided by MDCA. An example of this is the additional Image Processing Unit (IPSU) and High Bit Depth/Multispectral (HiBMs) camera that FLEX2 requires for its operations. CIR will also be responsible for the negotiation of the up-mass of replacement components for MDCA hardware and consumable items that are supplied by CIR as specified in CIR-MDCA Integration Agreement (FCF-IA-MV-MDCA). The consumable items required by FLEX2 include an adsorber cartridge and gas supply bottles. The safety assessments for these CIR items

5.0 MDCA STRUCTURAL/MECHANICAL

5.1 Chamber Insert Assembly

The MDCA CIA (See FIGURE 3 and FIGURE 4) is a multi-functional apparatus designed as the major platform used for mounting integral sub-system hardware to perform combustion science. The CIA consists of the Internal Apparatus, which is the functional hardware of the experiment, the Experiment Mounting Structure (EMS), upon which the Internal Apparatus is structurally based, and electrical and thermal interconnects. The Internal Apparatus is capable of dispensing, deploying and igniting fuel droplets with small velocities relative to the field of view of the diagnostic equipment. The Internal Apparatus of MDCA consists of two fuel dispensing systems, dual needle deployment mechanism, two hot wire igniters, an indexing fiber support mechanism, radiometers, pressure-sensitive igniter switch, and thermistors.

During the mission, the crew may replace igniters, the fiber support mechanism, fuel reservoir, and deployment needles.

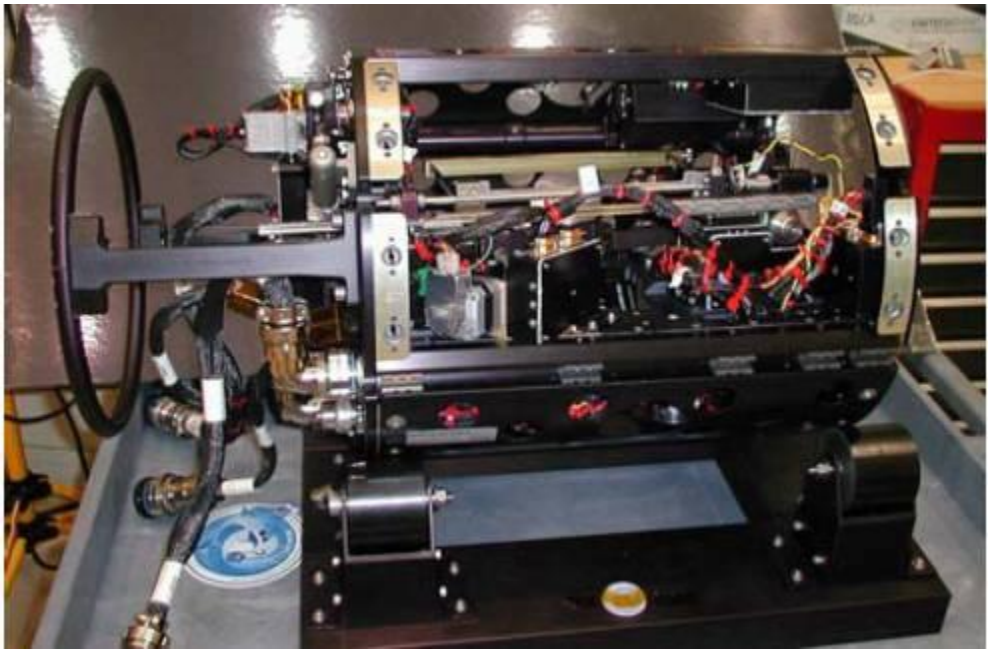


FIGURE 3 MDCA WITH SHROUDS REMOVED

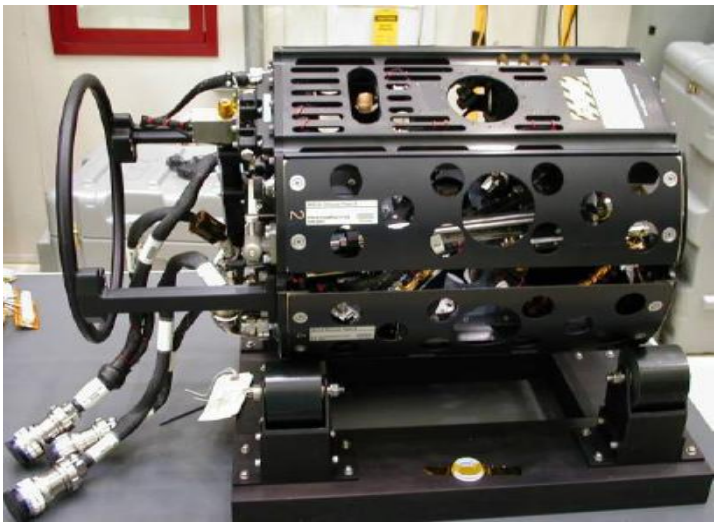


FIGURE 4 MDCA WITH SHROUDS

5.2 Internal Apparatus

5.2.1 Fuel Dispensing System

There are two fuel dispenser systems on the CIA. Each fuel dispenser system has two subassemblies; the replaceable fuel reservoir assembly and the dispenser motor assembly. The fuel reservoir is composed of a syringe and attached latching solenoid valve, an aluminum housing, and a transparent plate of Lexan F2100. The dispenser motor assembly is composed of a drive screw, a stepper motor, an incremental encoder, and structural base. During the software controlled dispensing sequence, fuel is dispensed from one syringe, through the latching solenoid valve to its own deployment needle. The solenoid valve is sent a voltage pulse that switches the valve to its alternate state. The syringe valve will be closed immediately following the dispensing of the fuel droplet.

Two identical dispenser systems are present for improved reliability when using a single type of fuel. Alternatively, each syringe may contain a mixture of fuels. In addition, the entire fuel reservoir assembly may be replaced if necessary by separating the fuel line bayonet, the valve power connectors, and loosening the captive fasteners. To monitor the temperature of the fuel, each syringe is outfitted with a thermistor.

FLEX2 requires the use of Ethanol, Decane, and a set of fuel mixtures. Additional 5 ml fuel reservoirs will be delivered in the FLEX2 cache of hardware, pre-filled with the

ratios and amounts of fuels required by the FLEX2 Science Requirements Document and combustion analysis. The table below gives an estimate of the mixtures, pending the results of a detonation analysis, and JSC toxicity assessment.

TABLE 2 FUELS FOR FLEX AND FLEX2 (5 ML RESERVOIRS)

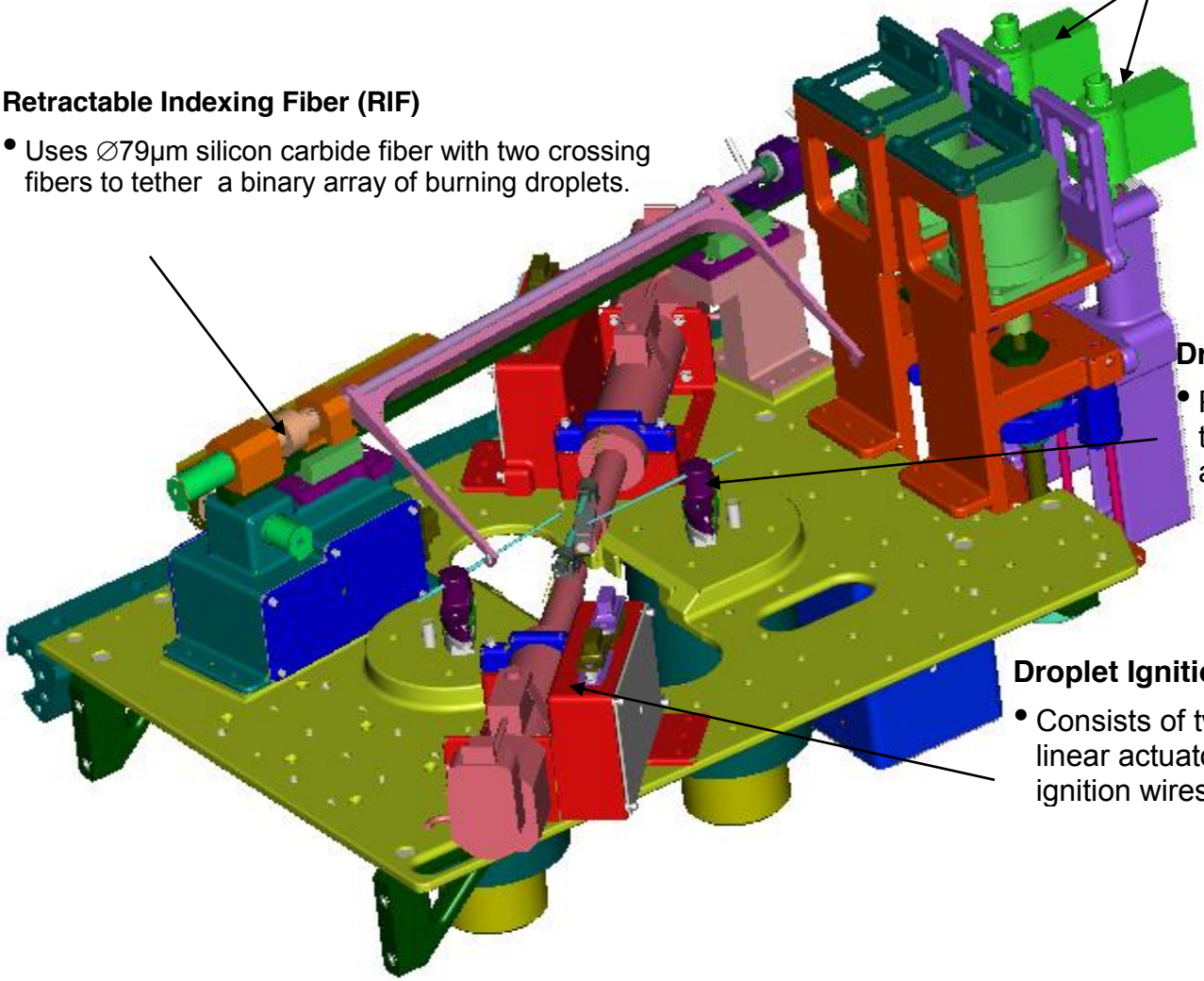
FLEX Fuels	FLEX2 Fuels	Mixture Ratio (%)
Methanol	Ethanol	100
Heptane	Decane	100
	Heptane/Hexadecane	10/90, 80/20, 85/15, 90/10, 95/5
	Propanol/Glycerol	10/90, 80/20, 85/15, 95/5…, 90/10
	Octane	100
	Heptane/Octane	10/90, 25/75, 40/60, 50/50, 55/45, 70/30, 85/15
	Decane/Propylbenzene	10/90, 25/75, 40/60, 55/45, 70/30, 85/15

Dual Fuel Dispenser System

- Dispenses fuel individually or simultaneously

Retractable Indexing Fiber (RIF)

- Uses ∅79µm silicon carbide fiber with two crossing fibers to tether a binary array of burning droplets.



Droplet Deployment System

- Precisely aligns two needles during the fuel droplet formation process and deployment sequence

Droplet Ignition System

- Consists of two opposing independently controlled linear actuator assemblies with two dual coil ignition wires for igniting droplet arrays.

FIGURE 5 MDCA/FLEX2 INTERNAL APPARATUS

5.2.2 Fuel Reservoir System

While the fuels loaded into the reservoirs differ between FLEX and FLEX2, the fuel reservoir design is identical. The fuel reservoir is a gas-tight glass syringe and stainless steel plunger. The plunger tip is covered by Teflon, which forms a dual wiper seal against the inside of the syringe. The syringe is covered by a tygon sleeve that is epoxied to the syringe at its top. The tygon-enshrouded syringe body is lightly clamped between a transparent cover and the housing. The syringe feeds fuel directly to an attached latching solenoid valve. It is also epoxied to the solenoid valve at its bottom. The fuel is fed from the valve through polytetrafluoroethylene (PTFE) tubing to the dispenser needles. A quick disconnect in the tubing is necessary to allow removal of the fuel reservoir.

Each syringe has two levels of containment. In addition to the tygon sleeve, the syringe is enveloped in an aluminum sleeve that is covered by a transparent Lexan 9604 barrier to protect against impact. See FIGURE 6.

Each syringe can contain 5.0 ml of fuel. Nearly all of the fuel in the syringe is capable of being dispensed.



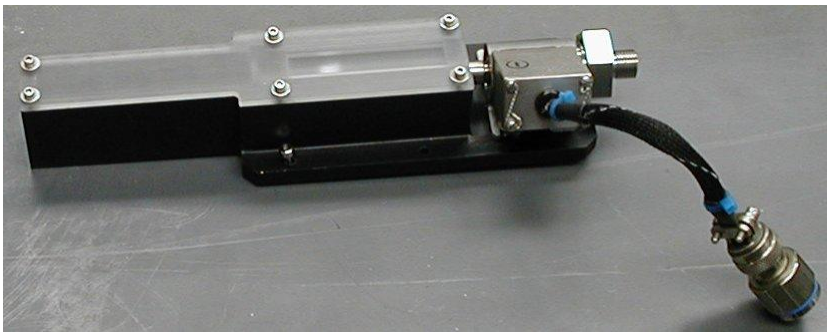


FIGURE 6 FUEL RESERVOIR



FIGURE 7 LEFT AND RIGHT FUEL RESERVOIR ASSEMBLY

5.2.3 Deployment System

The FLEX2 deployment mechanism is identical to that of MDCA/FLEX, and consists of two replaceable deployment needles mounted on brushless servo motors and bi-stable brakes. The deployment needles are flared tubes (See FIGURE 8) connected to the Dispenser System via .012 inch OD, .01 inch ID, PTFE tubing. The deployment needles are inverted relative to each other and opposed 180 degrees as illustrated in FIGURE 9. During deployment, the needles move in a plane common to the igniters and fiber. Each needle is moved by a brushless servo motor, which rotates the needles to a pre-determined position verified by motor encoder feedback. Just before dispensing fuel, the needles are positioned in close proximity to each other. Fuel from each of the fuel dispenser systems is routed to its dedicated deployment needle tip to create a single droplet of a desired size. Upon completion of fuel dispensing, the deployment motors actuate again, rotating to increase the gap between the needles and to minimize the needle surface contact with the droplet. Finally, the needles are rapidly withdrawn simultaneously. The rapid synchronized retraction of the needles causes the fuel droplet to detach from the deployment needles with little or no residual motion. Either deployment needle may be used to dispense the droplet via its fuel dispenser system. The needles may be replaced by separating the bayonet assembly in the fuel line and unfastening the needle fixture on the motor axis. Per the replacement procedure, which is performed at the ISS provided work bench, the deployment needles will be removed only after its cover has been installed. This cover, shown in FIGURE 10, prevents contact with the needle during removal/installation. Each needle has its own cover that remains attached during stowage. Needles will be individually bagged upon replacement to contain any residual fuels and broken material.

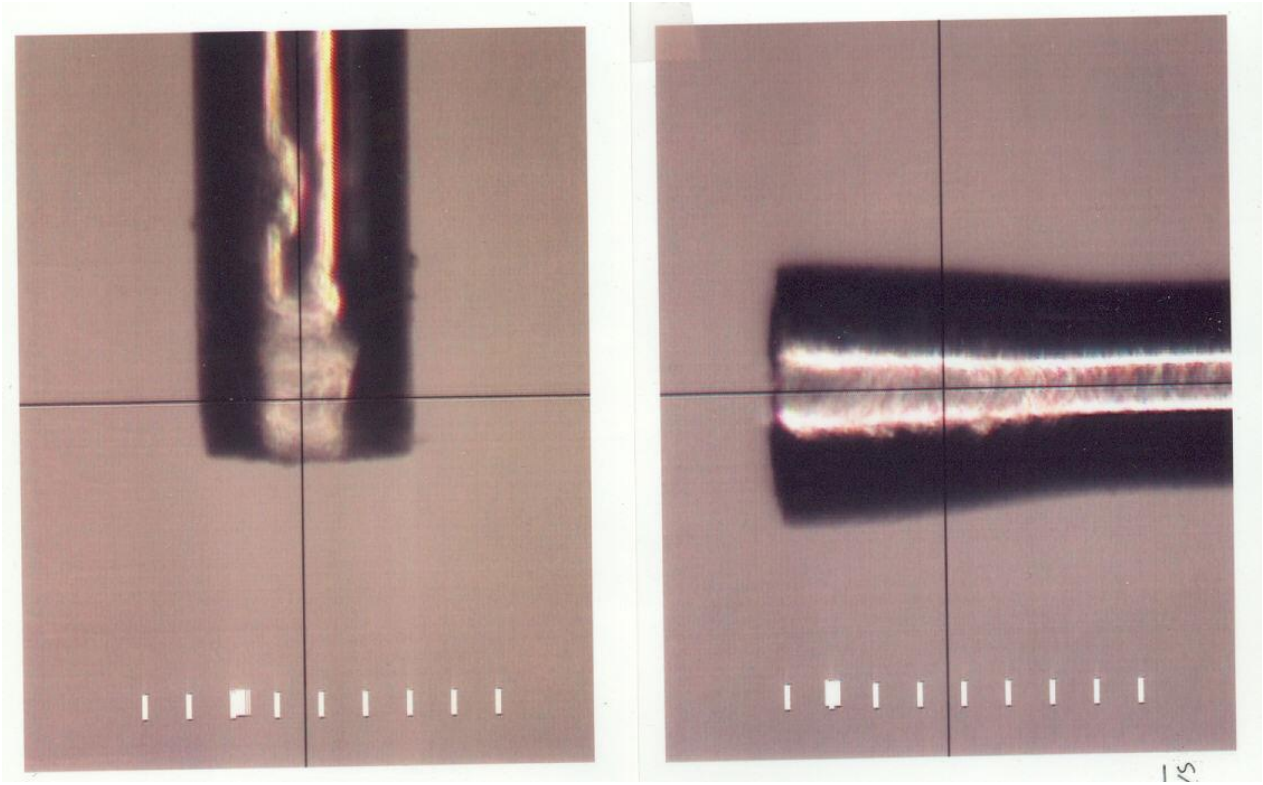


FIGURE 8 NEEDLE TIP BEFORE FLARING (LEFT) AND AFTER FLARING (RIGHT)

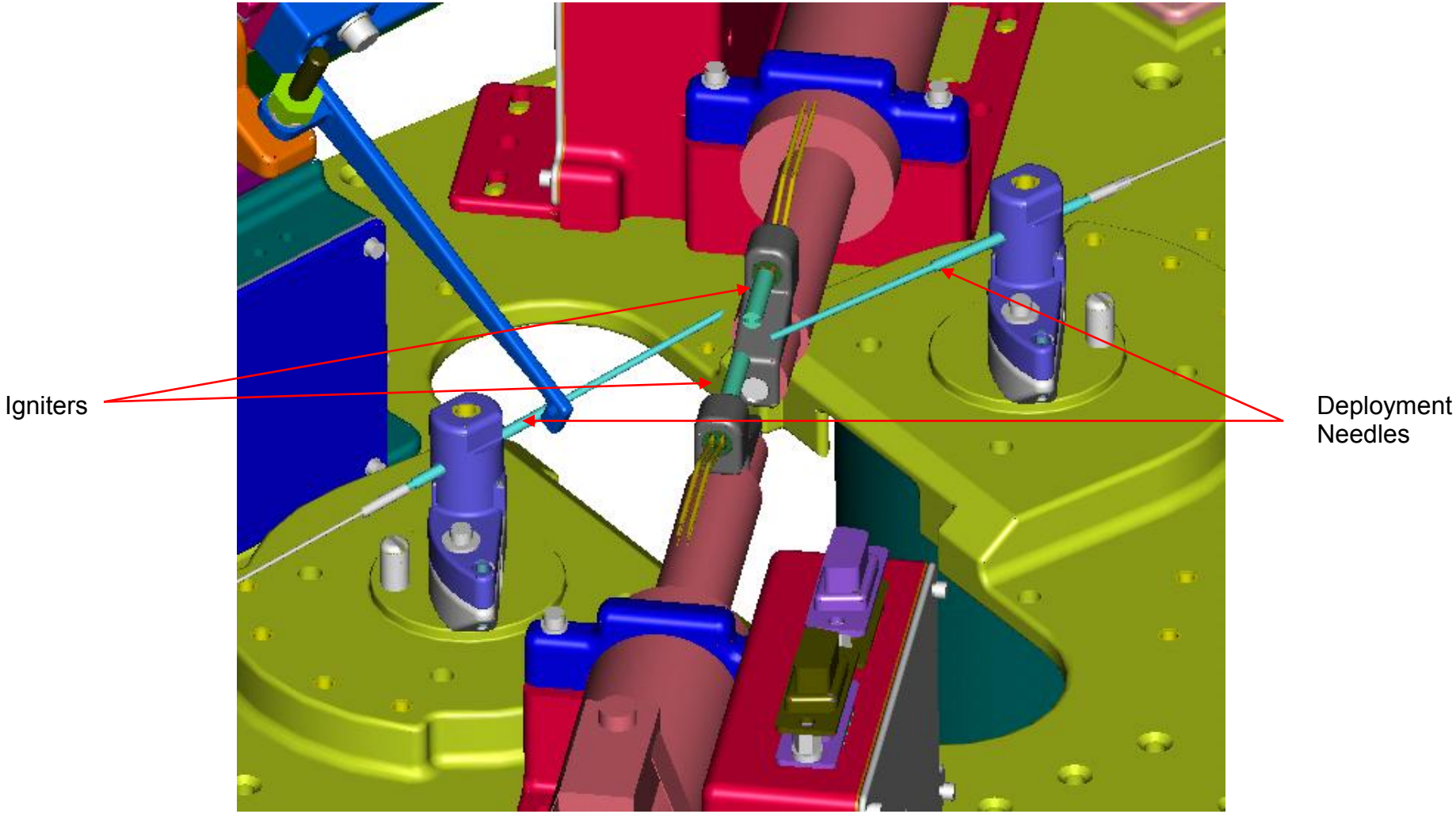


FIGURE 9 DROPLET DEPLOYMENT AND IGNITION AREA

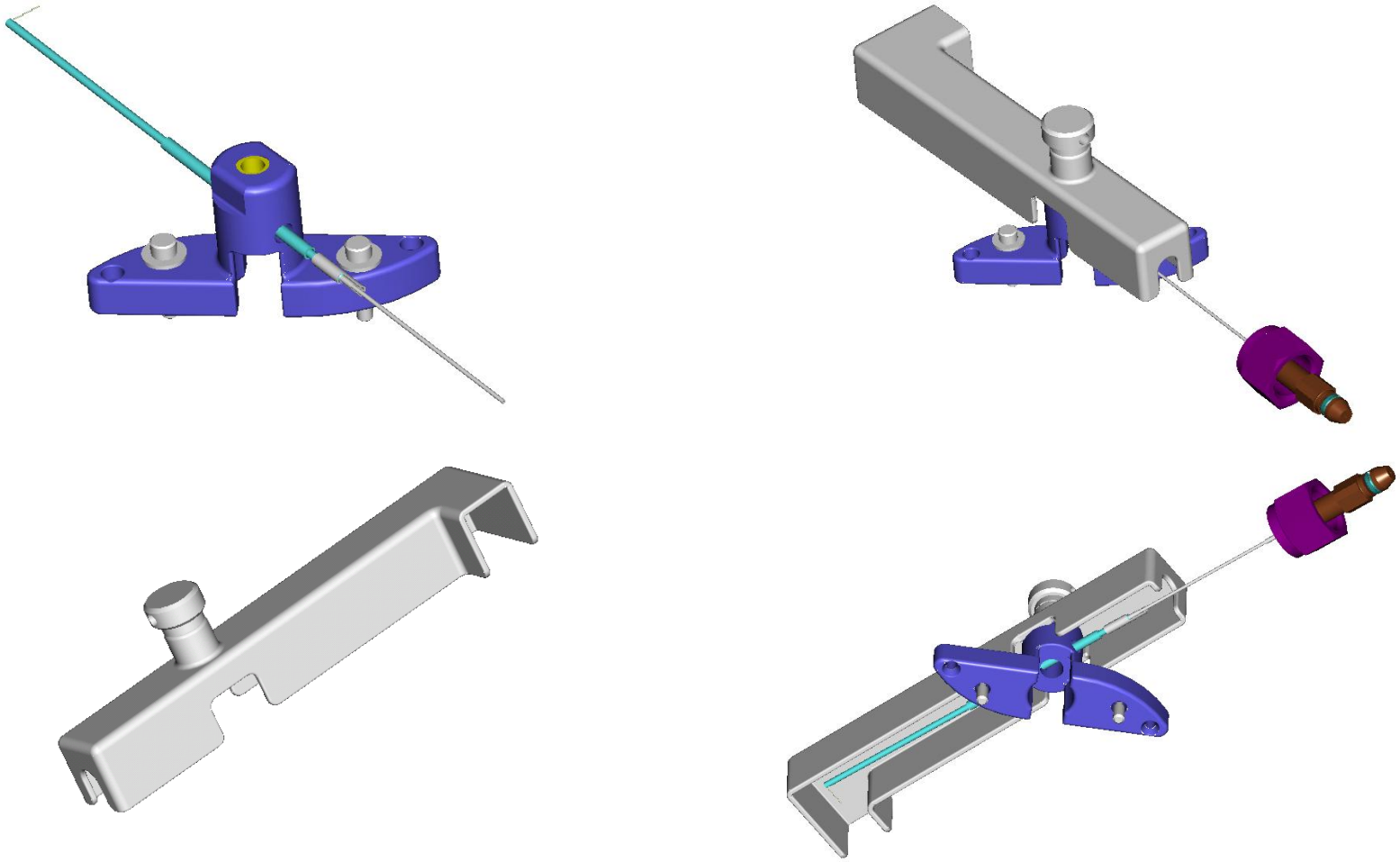
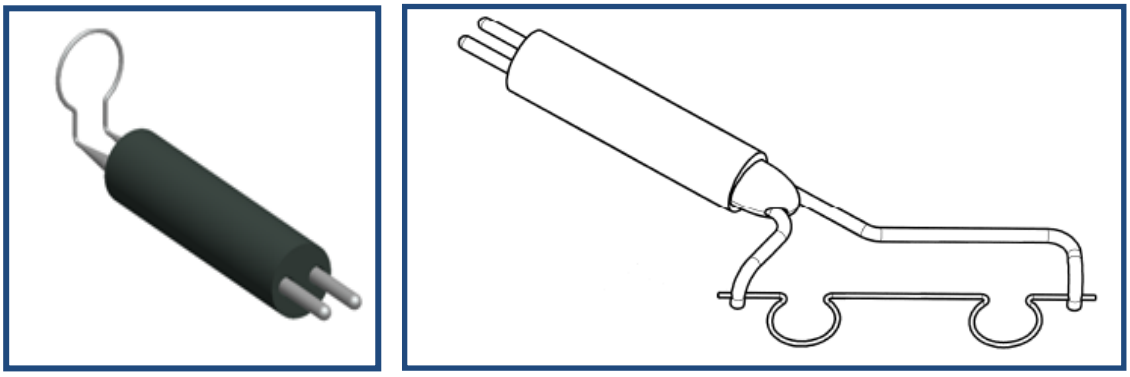


FIGURE 10 VIEWS OF NEEDLE ASSEMBLY AND NEEDLE COVER

5.2.4 Ignition System

While the MDCA/FLEX igniter tips uses two identical round heating elements with single loops, the FLEX2 ignition tips consist of two heating elements with a dual loop configuration. The elements are Kanthal hot wire igniters mounted on removable ceramic mounts (See FIGURE 11). The elements have can simultaneously ignite a two bead droplet array on the retractable index fiber. The ceramic mount is mounted to a bracket on the drive shaft. Each drive shaft is driven by a rotary stepper motor and an enclosed drive screw. Each hot wire igniter is positioned opposite the other. The rotary stepper motor moves the igniter linearly into the deployment area to a predetermined position just before dispensing of the droplet. Motor encoders automatically verify the igniter position. The fuel is dispensed, the droplet is deployed, and a wait period passes to allow some fuel vaporization. Then both igniters are pulsed with a brief current to heat the igniter and ignite the fuel vapor surrounding the droplet. Finally, the igniters rapidly retract out of the field of view of the diagnostic equipment. Each igniter may be replaced by pulling the ceramic mount from the bracket. The electrical connection is made when the ceramic mount is plugged onto the bracket.

The igniter tips will be stowed in bags. The crew will be required to wear gloves while performing this procedure. Additionally, during this operation, which is performed at the ISS provided work bench, the covers for the deployment needles will be installed to prevent inadvertent contact.



FLEX FLEX2

FIGURE 11 FLEX AND FLEX2 IGNITER TIPS

5.2.5 Fiber Support System

The FLEX2 fiber support mechanism is based on that of MDCA/FLEX, with the addition of crossing fibers for tethering an array of droplets. The FLEX2 experiments employ a set of 79 micron diameter, 229 mm long silicon carbon-carbide fibers with an additional set of crossing fiber pairs of the same material to suspend fuel droplets in an array of predetermined dimension. The main fibers are supported at both ends by the arms of a U-shaped bracket. The lower portion of the U- bracket is mounted to an axle. The axle is attached to a rotary stepper motor that rotates the bracket and fiber

approximately 120 degrees about the axle so that the fiber is in the center of the fuel deployment area. Retractable Indexing Fiber (RIF) Sub-Assembly design features replaceable Fiber Arm, both rotational and linear over travel hard stops. Stepper Motors are protected by use of slip clutches. See FIGURE 12.

The droplet may be deployed at various intersections of the crossing fiber pairs with the main fiber. There are 3 pairs of crossing fibers with two pairs 12mm apart, and 1 pair at 24mm apart. They are attached by epoxy to a cross member that is attached to the U-shaped fiber arm bracket. A second rotary stepper motor actuates a rack and pinion mechanism. The rack and pinion mechanism allows the fiber to be indexed along the axis of the fiber. Deployment onto the fiber is limited to the central 129 mm of the fiber because using the extreme ends of the fiber would bring the bracket arms close enough to interfere with the combustion event.

When a tethered droplet array is desired, one of the crossing fibers is moved into position, the droplet is dispensed onto the cross fiber, and the fiber is translated the distance of the cross fiber pair spacing. There are two crossing fiber pairs spaced 12mm apart, and one pair spaced 24mm apart. The next droplet is then dispensed onto the second cross fiber pair. The deployment needles retract, and the igniters ignite the droplets.

The hardware-restricted spherical volume around the center of the droplet is called the zone of exclusion. When the fiber is not being used, the fiber will be in the retracted mode, clear of the deployment area and diagnostic fields of view. Rotational retraction speeds of the RIF are nominally from 2°-6°/sec.

The fiber support mechanism may be replaced in case of damage with stowed spares by unfastening the captive fasteners shown in FIGURE 13. The Fiber Arm is replaced at the ISS provided work bench after the CIA is removed from the chamber. It requires no complicated alignment techniques and no electrical power. Included in the Fiber Arm replacement procedure is the installation of the protective cover for the fiber which prevents inadvertent contact. The Fiber Arm is stowed with this cover installed. To prevent handling damage, each Fiber Arm Assembly is shipped with a protective cover (FIGURE 14).

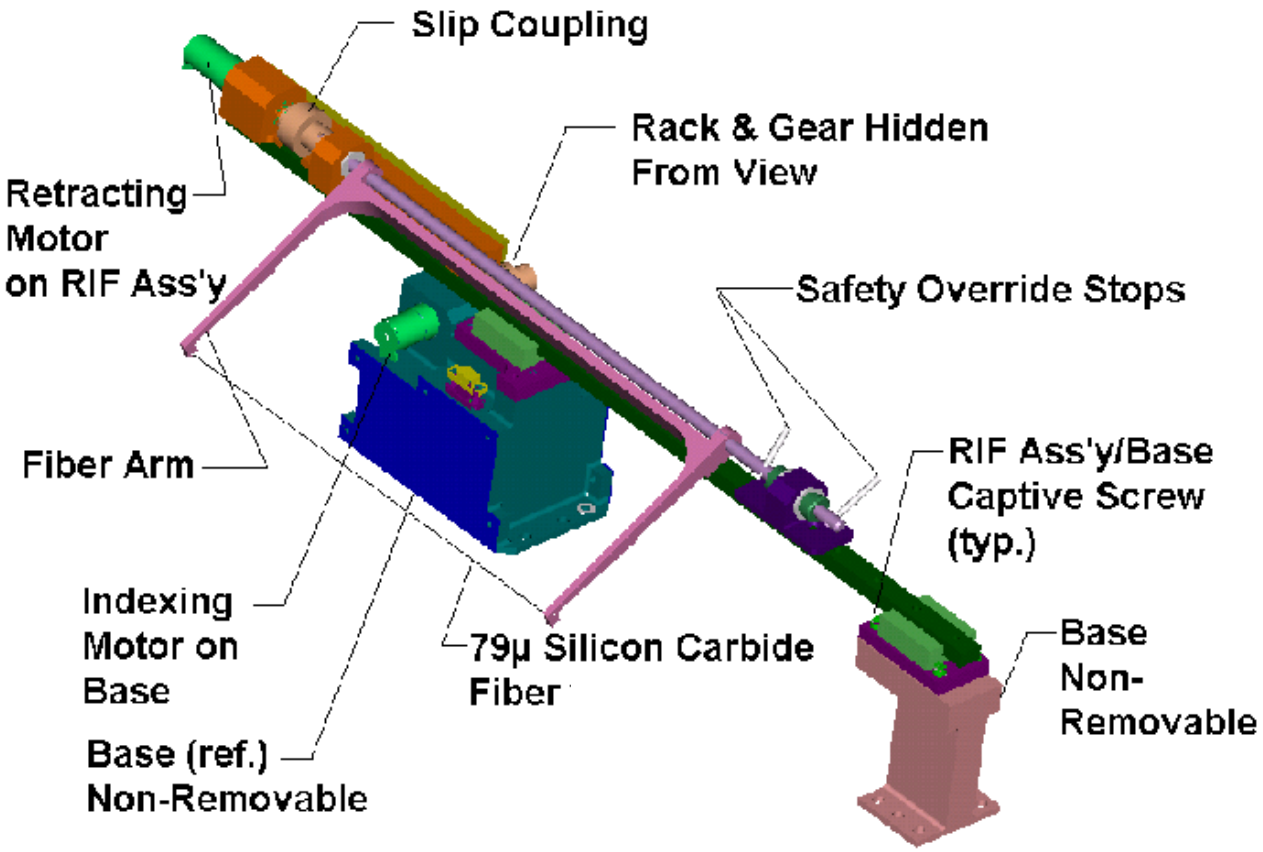


FIGURE 12 RIF ASSEMBLY WITH FLEX FIBER ARM

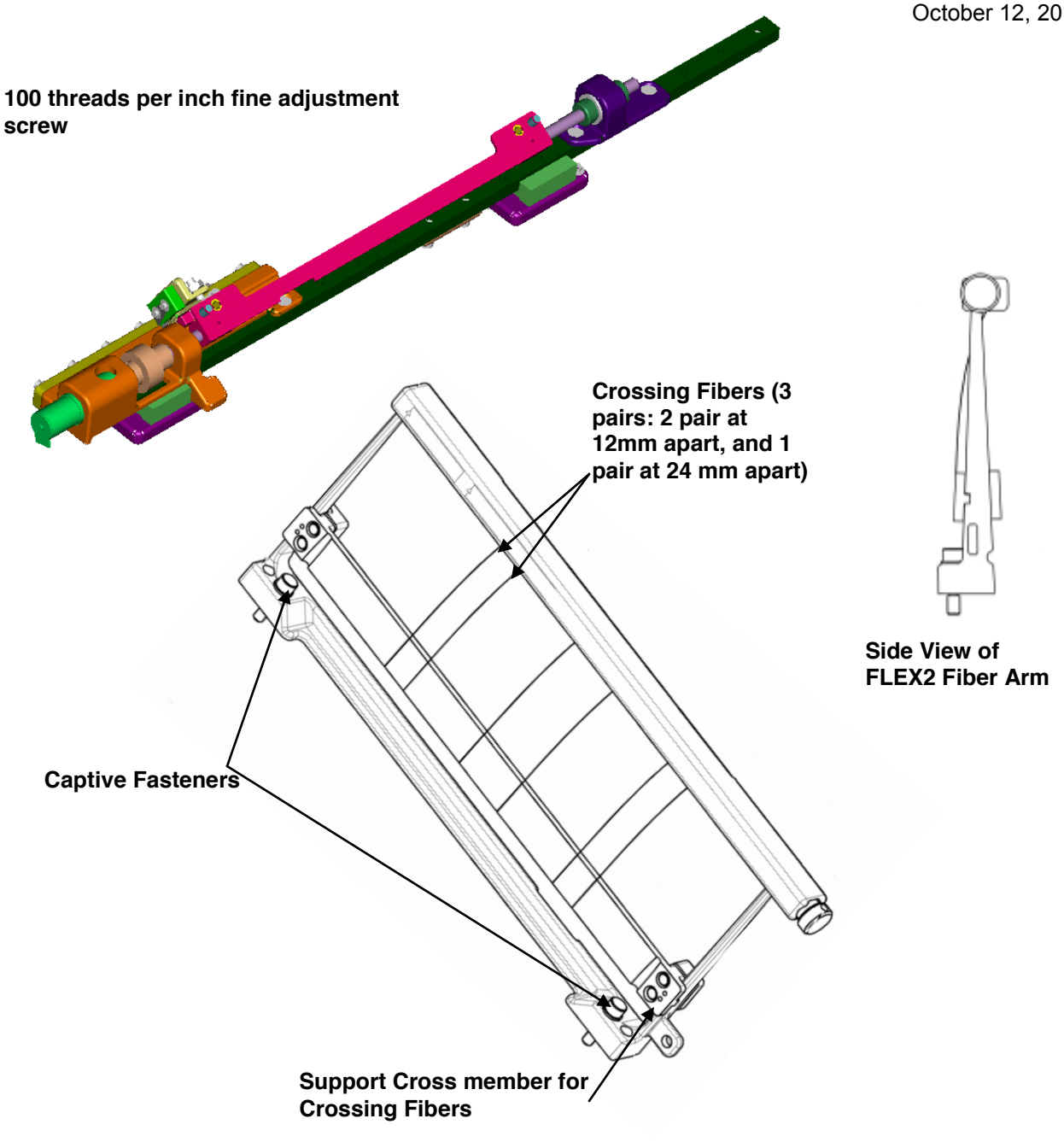


FIGURE 13 RIF SUB-ASSEMBLY & FLEX2 CROSSING FIBERS FIBER ARM

In accordance with crew recommendations, the Replaceable Fiber Arm and Fiber Arm Protective cover have been designed for ease of installation and removal. Hand adjustment of a captive screw at end of the cover is used to secure the cover to the Fiber Arm. These covers remain assembled to the Fiber Arms at all times when not installed on the RIF.

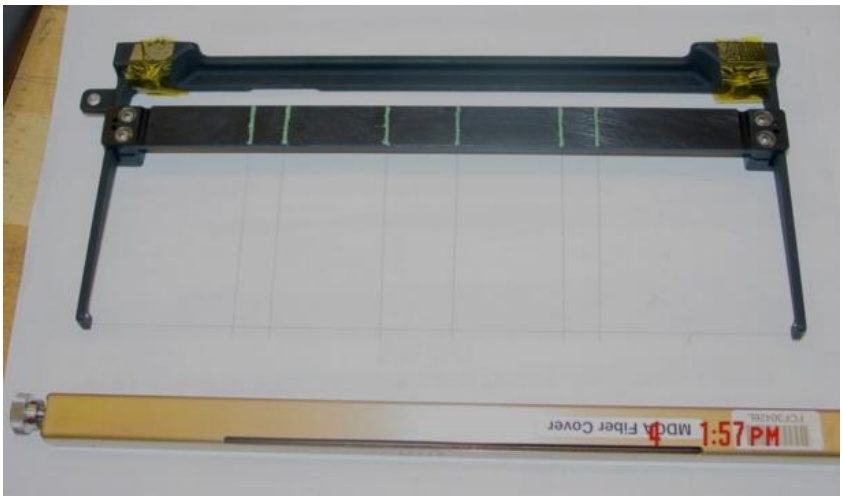


FIGURE 14 FLEX2 FIBER ARM AND PROTECTIVE COVER

5.2.6 Water Cooling System

The details of the water cooling system are unchanged by the FLEX2 operations, and are included in the MDCA safety data package MDC-DOC-1679. The CIA will take in water from the CIR Water Thermal Control System through the IRR. CIA intake and exhaust lines will connect to the IRR. The CIA water lines provide the interface between the CIR IRR and the MDCA CIA structure. The coolant system will connect to the CIR IRR with quick disconnects.

5.2.7 Pressure Sensitive Igniter Switch

The details of operation of the pressure sensitive igniter switch are unchanged by the FLEX2 operations, and are included in the MDCA safety data package MDC-DOC-1679. While the physical and electrical operation of the switch is unchanged, its role no longer functions as a hazard control in FLEX2. The updated detonation analysis shows that the fuel to O2 mixture ratio drops below the flammability limit as pressure increases. Therefore no hazard is associated with high pressure operations.

6.0 MDCA ELECTRICAL/AVIONICS

6.1 MDCA Avionics Package Hardware

The MDCA Avionics Package is made of aluminum alloy, 6061-T6. Some structural components may be composed of corrosion resistant stainless steel, A286.

The Avionics Package is an aluminum box that provides a physical housing, an EMI shield, and connection to the CIR Optics Bench electrical interconnects. The housing will be mounted on the Optics Bench at the Principal Investigator Location (PIL). This housing has inlet and exhaust vents to interface with the CIR Air Thermal Control System. The housing is illustrated in FIGURE 15 and FIGURE 16.

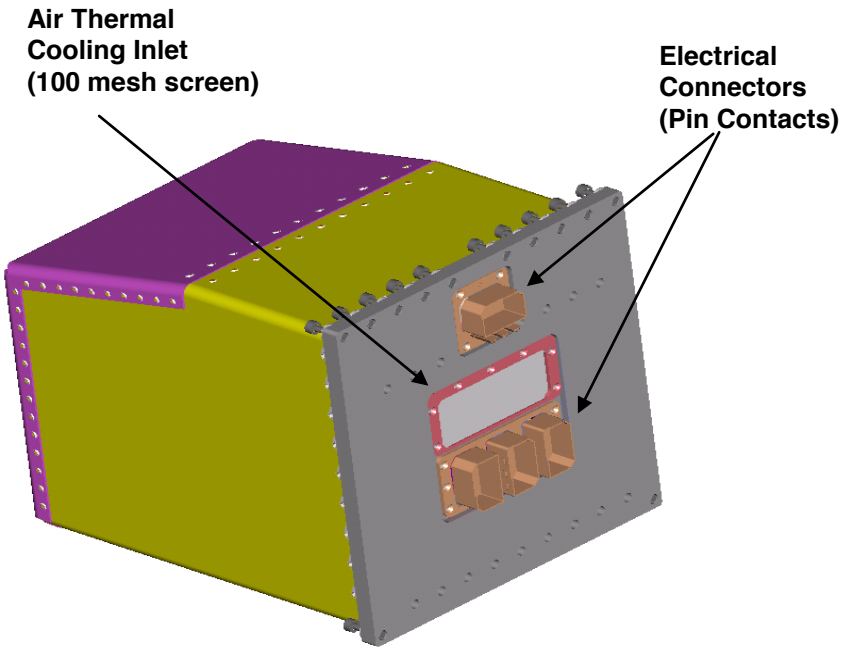


FIGURE 15 MDCA AVIONICS PACKAGE (BOTTOM)

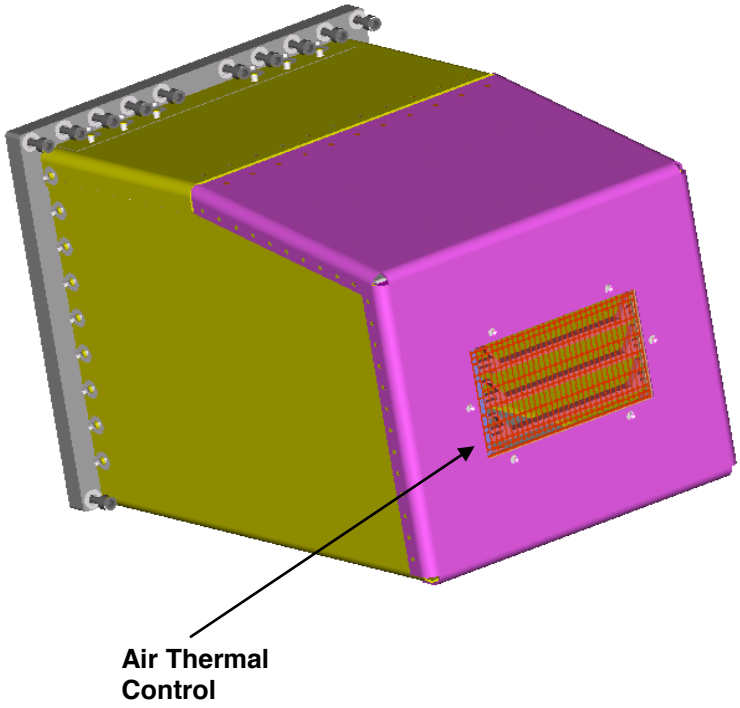


FIGURE 16 MDCA AVIONICS PACKAGE (TOP)

9.3 MDCA/FLEX2 Test Operations

9.3.1 Number of Test Points

FLEX2 has a test matrix with 550 points, as enumerated in Table 3 of the Science Requirements Document (SRD). The test points are distributed among mixed fuel tests and pure fuel tests. The pure fuel tests include the fiber droplet array tests and the flow velocity tests, created by the translation of the fiber during the burn event. A preliminary test matrix was constructed to address these requirements, and to keep the number of experiments to a manageable level, resulting in 456 test points.

9.3.2 Resource Estimates

This section presents the preliminary estimated amount of necessary resources determined from the experiment science requirements and the preliminary test matrix. The fuel reservoirs have a 5 ml capacity, but the actual volumes are dependent on the results of future detonation analysis.

FLEX2 requires the following items based on 456 test points:

- 23 - 3.8 liter Bottles
  - 9 - 3.8 liter 30%O2 70%N2 bottles
  - 7 - 3.8 liter 30%O2 70%He bottles
  - 7 100%He bottles
- 22 Fuel Reservoirs
  - 2 Heptane/Hexadecane
  - 2 Propanol/Glycerol
  - 6 Heptane/Octane
  - 6 Decane/Propylbenzene
  - 3 Decane
  - 3 Ethanol
- 1 Large Adsorber Cartridge
- 2 Fiber Arms

9.3.3 Chamber Environments

The following matrices show the chamber atmospheres, pressures, fuels, droplet sizes, and flow velocities for the FLEX2 experiments.

TABLE 3 FLEX2 PRELIMINARY TEST MATRIX FOR PURE FUELS

FLEX-2 Pure Fuels									
Investigation Class	States	Matrix Variables						Permutations	
		Drop Diam	O2 %	Dil	Bead Spacing	Pressure	Flow Velocity		
1- decane - flow	States	2	4	2		1	3		48
	Matrix Values	3 mm 4 mm	17% 21% 25% 30%	N2 He		1 atm	0.1 cm/sec 0.3 cm/sec 0.5 cm/sec		
1 - ethanol - flow	States	2	4	2		1	3		48
	Matrix Values	3 mm 4 mm	17% 21% 25% 30%	N2 He		1 atm	0.1 cm/sec 0.3 cm/sec 0.5 cm/sec		
1- decane - quiescent	States	2	4	2		3	-		48
	Matrix Values	3 mm 5 mm	17% 21% 25% 30%	N2 He		0.5 atm 1.0 atm 2.0 arm	-		
1 -ethanol -quiescent	States	2	4	2		3	-		48
	Matrix Values	3 mm 5 mm	17% 21% 25% 30%	N2 He		0.5 atm 1.0 atm 2.0 arm	-		
1 - decane - array	States	1	3	2	2	2	-		24
	Matrix Values	3 mm	17% 21% 25%	N2 He	L1 L2	0.5 atm 1.0 arm	-		
1 - ethanol - array	States	1	3	2	2	2	-		24
	Matrix Values	3 mm	17% 21% 25%	N2 He	L1 L2	0.5 atm 1.0 arm	-		
Pure Fuel Sub-total								***0***	***240***

TABLE 4 FLEX2 PRELIMINARY TEST MATRIX FOR MIXED FUELS

FLEX-2 Mixed Fuels								
Investigation Class	States	Matrix Variables					Permutations	
		Drop Diam	O2 %	Dil	Pressure	Fuel Mix		
1- heptane/hexadecane	States	2	3	1	3	2		36
	Matrix Values	3 mm 4 mm	17% 21% 30%	He	0.7 atm 1.0 atm 2.5 arm	F1 F2		
1- propanol/glycerol	States	2	3	1	3	2		36
	Matrix Values	3 mm 4 mm	17% 21% 30%	He	0.7 atm 1.0 atm 2.5 arm	F1 F2		
1- heptane/octane	States	1	3	2	2	6		72
	Matrix Values	3 mm	17% 21% 30%	N2 He	1.0 atm 2.0 arm	10/90 25/75 40/60 55/45 70/30 8515		
1- decane/propybenzene	States	1	3	2	2	6		72
	Matrix Values	3 mm	17% 21% 30%	N2 He	1.0 atm 2.0 arm	10/90 25/75 40/60 55/45 70/30 8515		
Mixed Fuel Sub-total							***0***	***216***